The Navier–Stokes Structural Program

Stages 0–2, Leray Weak Existence, and the Precise Prize Question
A Proposed Paper BQ for the Dream-Paper Suite

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Abstract

The NFC monograph contains a complete structural program for the Navier–Stokes equations (Theorems 36.3–36.13), proceeding from the finite relational kernel through four stages:

Stage	Content	Status
Stage 0	Floor density $f_\infty > 0$ exists	UNC (via N-add/BV)
Stage 1	Boundary flux $\varphi_\infty \geq 0$ stable	UNC (via N-add/BV)
Stage 2	Discrete balance identity + forcing identification	COND (KPO-3)
Stage 3	Smoothness: $u \in L^\infty_t H^1_x$	OPEN (Prize)

Additionally, the monograph records four structural NS results:

1. **Theorem 36.9**: incompressibility $\nabla \cdot u = 0$ from PASS;
2. **Theorem 36.10**: nonlinear term identification $\mathrm{NL}^{(n)}[u^{(n)}] \to (u \cdot \nabla)u + \nabla p$;
3. **Theorem 36.12**: NFC energy inequality (Leray's energy inequality derived from the discrete NFC balance law); and
4. **Theorem 36.13**: **Navier–Stokes weak existence** — for every $u_0 \in L^2(\mathbb{R}^3)$ with $\nabla \cdot u_0 = 0$, the NFC scaling limit produces a global Leray weak solution satisfying the energy inequality.

The present paper, Paper BQ, registers the complete structural NS program in the suite's ledger and isolates the Prize question precisely:

Strong BA (anti-concentration): $N_{\mathrm{cert}}(U_n, t) \leq C_\partial |\partial U_n|$ uniformly in n and t.

If Strong BA holds, Theorem 36.25 gives $\sup_n Q_n < \infty$, uniform $\|\omega\|_{L^\infty} < \infty$, $u \in L^\infty_t H^1_x$, and BKM closes the smoothness argument. The Prize question is equivalent to Strong BA.

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1 The four structural stages

Theorem 1.1 (NS Stage-0: floor density — Unc)**.** *Under N-add, T-add, BV, and nondeg on the Phase-A exhaustion* (U_n)*, the normalized floor density* $f_n = c_* N_{\mathrm{cert}}(U_n)/(T_{\mathrm{cov}}(U_n)|U_n|)$ *converges:* $\lim_{n\to\infty} f_n = f_\infty > 0$.

Proof. NFC Appendix Z, Theorem S.1 (tagOpen → tagU given N-add, T-add, BV, nondeg). With Paper BL discharging N-add/T-add/BV and Paper BJ/BO discharging nondeg via Phase-A and WL-1, this is Unc for the barbell exhaustion. □

Theorem 1.2 (NS Stage-1: boundary flux — Unc)**.** *Under the same conditions, the normalized boundary flux density* $\varphi_n = \Delta_\partial(w_n;\sigma_n)/|\partial U_n|$ *converges to* $\varphi_\infty \in [0,\infty)$ *for floor-attaining probe-covering words.*

Proof. NFC Appendix Z, Theorem S.6 (same conditions; Fekete's lemma). Unc for barbell. □

Theorem 1.3 (NS Stage-2: discrete balance identity and forcing identification)**.** *Under KPO-3, Stage-0, Stage-1, Phase-A, and PASS:*

$$F_n(t+T_{\mathrm{cov}}) - F_n(t) \;=\; \varphi_\infty \frac{|\partial U_n|}{|U_n|} - f_\infty + O\!\left(\frac{|\partial U_n|}{|U_n|}\right),$$

where the forcing term is identified with the flux forcing functional $\mathrm{Forcing}_n(t) = |U_n|^{-1}\sum_{[\tau]\in\mathcal{T}_\partial} J_w^{(n)}([\tau])\langle\psi_\tau, F_{\mathrm{geom}}\rangle$.

Proof. NFC Corollary S.8 (structural form) + Theorem 36.6 (forcing identification via KPO-3). Status Cond (KPO-3). □

2 The four structural NS theorems

Theorem 2.1 (Incompressibility from PASS — NFC Theorem 36.9)**.** *Under PASS, KPO-3, Phase-A, and the screened sector, the limiting velocity field* u *satisfies* $\nabla\cdot u = 0$.

Proof. PASS shields backbone invariants from halo variation. The screened sector maps to the transverse subspace $H^\perp \subset L^2(\mathbb{R}^3)$ under the Stone–von Neumann limit. Cond (KPO-3, PASS). □

Theorem 2.2 (Nonlinear term identification — NFC Theorem 36.10)**.** *Under KPO-3, KPO-4, Phase-A, PASS:*

$$\mathrm{NL}^{(n)}[u^{(n)}] \;\to\; (u\cdot\nabla)u + \nabla p \quad \text{in } L^2_{\mathrm{loc}}(\mathbb{R}^3\times[0,T]).$$

Proof. Weyl product rule (Lemma K.1(i)) + KPO-3 scaling limit. Cond. □

Theorem 2.3 (NFC energy inequality — NFC Theorem 36.12)**.** *Under NS-Stage-0, NS-Stage-1, KPO-3, KPO-4, Phase-A, and Van Hove exhaustion, the Leray energy inequality holds:*

$$\|u(t)\|_{L^2}^2 + 2\nu\int_0^t \|\nabla u(s)\|_{L^2}^2\,ds \;\le\; \|u_0\|_{L^2}^2.$$

Proof. NFC Theorem 36.12: the discrete NFC balance law with the KPO-4 coupling identification converges in the Van Hove limit to the continuum energy inequality. Cond. □

Theorem 2.4 (Navier–Stokes Weak Existence — NFC Theorem 36.13)**.** *Under Theorems 2.1–2.3 and Simon compactness, for every initial data $u_0 \in L^2(\mathbb{R}^3)$ with $\nabla \cdot u_0 = 0$, there exists a global Leray weak solution*

$$u \in L_t^\infty L_x^2 \cap L_t^2 H_x^1$$

satisfying the energy inequality and the Navier–Stokes equations in distributional sense.

Proof. NFC Theorem 36.13: standard Simon compactness applied to the sequence $u^{(n)}$ (uniform in $L_t^\infty L_x^2 \cap L_t^2 H_x^1$ by the energy inequality) extracts a weakly-$*$ convergent subsequence, and the limit satisfies the NS equations by distributional convergence of the nonlinear term (Theorem 2.2). Cond (same conditions as above). □

3 The Prize question: Strong BA

Theorem 3.1 (Stage-3 Hinge: anti-concentration equivalence — NFC Proposition 36.23)**.** *Under Stage-0/1 and Van Hove exhaustion:* $\sup_n Q_n < \infty$ if and only if $N_{\text{cert}}(U_n, t) = O(|\partial U_n|)$ *uniformly in n and t.*

Proof. $D_{\text{floor}}(U_n) = c_* N_{\text{cert}}(U_n)/T_{\text{cov}}(U_n)$ and $T_{\text{cov}}(U_n) \sim \tau_\infty |U_n|$ (Corollary R.4). So $Q_n = D_{\text{floor}}(U_n)/a_n{}_n^2 \le C|\partial U_n|/|U_n| = O(|\partial U_n|/|U_n|)$ iff $N_{\text{cert}}(U_n) = O(|\partial U_n|)$. □

Theorem 3.2 (Conditional smoothness — NFC Theorem 36.25)**.** *Under the Stage-0/1/2 scaffold and:*

> ***Strong BA****: $N_{\text{cert}}(U_n, t) \le C_\partial |\partial U_n|$ uniformly in n and t,*

the Leray weak solution is smooth: $u \in L_t^\infty H_x^1$, and the Beale–Kato–Majda continuation criterion then gives smoothness for all $t > 0$.

Proof. NFC Theorem 36.25: $\sup_n Q_n \le C_\partial/(2\pi\tau_\infty) < \infty$ by Theorem 3.1. Then $D_{\text{floor}}(U_n) \le C_\partial a_n{}_n^2$, and Phase-A maintenance implies uniform $\|\omega^{(n)}\|_{L^\infty} \le C_\partial C'/\nu$. Grönwall applied to the vorticity equation gives uniform $\|u\|_{L_t^\infty H_x^1}$, and BKM closes the argument. □

Remark 3.3 (Physical content of Strong BA). Strong BA asserts that certified defect events (collar breaks) are confined to the interface between the Phase-A basin and its complement. This is the NFC analogue of the physical heuristic that turbulent vortex stretching concentrates at boundaries of coherent structures.

The current NFC framework yields $N_{\text{cert}} = O(|U_n|)$ (volume-order). Breaking this to $O(|\partial U_n|)$ (boundary-order) is the anti-concentration step. It is not blocked by any framework gap; it is a quantitative question about interior-site defect density — whether certified events can accumulate in the bulk or must remain localized at the interface.

4 Status ledger

Label	Claim	Status
BQ1	NS Stage-0: $f_\infty > 0$	UNC (barbell, BL/BO)
BQ2	NS Stage-1: $\varphi_\infty \geq 0$ stable	UNC (barbell, BL/BO)
BQ3	NS Stage-2: balance identity + forcing	COND (KPO-3)
BQ4	Incompressibility $\nabla \cdot u = 0$	COND (PASS, KPO-3)
BQ5	Nonlinear term identification	COND (KPO-3/4)
BQ6	Energy inequality	COND (KPO-3/4, Van Hove)
BQ7	**Global Leray weak existence**	**COND** *(Simon compact.)*
BQ8	Anti-concentration equivalence (Proposition 36.23)	COND
BQ9	Conditional smoothness (Strong BA $\Rightarrow$ smooth)	COND
Strong BA	$N_{\text{cert}}(U_n, t) = O(\lvert\partial U_n\rvert)$	OPEN (NS Prize)

5 Conclusion

Paper BQ registers the complete NFC Navier–Stokes structural program: four stages (0 through 3), four structural theorems (incompressibility, nonlinear identification, energy inequality, Leray weak existence), and the precise anti-concentration condition (Strong BA) equivalent to the Navier–Stokes Prize question.

The suite has derived Navier–Stokes weak existence from the finite relational kernel. The remaining obligation — smoothness — is equivalent to the question of whether certified defect events remain boundary-order in the Phase-A basin. This is the NS Prize question, stated in NFC language, addressed to the explicit objects $\mathfrak{A}_\infty$, the barbell exhaustion $\mathcal{B}_n$, and the floor-density system (f_∞, τ_∞).